

Incident Response Case Study – SOC Analyst Portfolio (TryHackMe + LetsDefend)

This project is based on simulated lab environments (TryHackMe and LetsDefend), focusing on incident response and threat investigation for portfolio purposes.

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Environment: TryHackMe (Linux/Windows Incident Surface) + Splunk Free (Ubuntu)

Case Type: Lab / Simulated Incident (Training)

Skills Demonstrated

- SIEM Investigation (Splunk)
- Incident Response
- Threat Hunting
- Windows Event Analysis
- Linux Host Analysis
- Log Analysis & Event Correlation
- IOC Identification
- MITRE ATT&CK Mapping
- Incident Documentation

SOC Investigation Case Study – Multi-Stage Compromise

Scenario

While reviewing security logs and endpoint activity within a controlled lab environment, I identified several indicators that suggested a potential compromise affecting both Windows and Linux systems.

The objective was to determine whether malicious activity had occurred, identify attacker techniques, and understand how the compromise was established.

Initial Observations

During the investigation, several suspicious indicators were identified:

- Failed authentication attempts
- PowerShell activity
- Persistence mechanisms
- Remote access software
- Privilege escalation indicators
- Suspicious file downloads

These indicators suggested that the systems may have been compromised through multiple attack stages rather than a single event.

Initial Hypothesis

Based on the available evidence, I suspected that an attacker had established access, deployed persistence mechanisms, and attempted to maintain long-term control of the environment.

To validate this hypothesis, I began a structured investigation using Splunk, Windows event logs, Sysmon telemetry, and Linux artifacts.

Investigation Objectives

- Identify the attacker's initial access method
- Determine persistence mechanisms used
- Investigate privilege escalation activity
- Review remote access tooling
- Validate evidence using SIEM telemetry
- Document findings and response actions

Investigation Phase 1 – Host Analysis

Objective

After identifying several suspicious indicators, I began reviewing both Windows and Linux hosts to determine whether persistence mechanisms or unauthorized access had been established.

Investigation Process

I reviewed local user accounts, scheduled tasks, startup locations, services, registry modifications, and system configurations.

The goal was to identify evidence of persistence, privilege escalation, or attacker-controlled access.

Key Findings

Unauthorized User Account

A suspicious local account named **attacker** was identified on the system.

The account appeared inconsistent with normal administrative activity and warranted further investigation.

Privilege Escalation

System configuration reviews revealed modifications to the sudoers file that granted elevated privileges.

This suggested an attempt to maintain administrative access within the environment.

Persistence Mechanisms

Multiple persistence techniques were identified, including:

- Cron job execution at startup
- Suspicious system service configuration
- Windows startup-related artifacts

These mechanisms could allow malicious activity to survive system reboots.

Remote Access Activity

Evidence of remote administration tooling was identified, including:

- AnyDesk
- Tunneling-related scripts

These findings suggested possible long-term remote access to the environment.

Credential Access Indicators

Registry modifications associated with credential access were identified.

Further analysis revealed WDigest-related configuration changes that may expose credentials in memory.

Additional Findings

Further host analysis identified a suspicious executable located in a temporary directory (/tmp/simple), an unauthorized software package installation, and a hidden NTFS volume.

Although these artifacts alone did not confirm malicious activity, they provided additional evidence that the environment had been intentionally modified and required further investigation.

Assessment

At this stage of the investigation, the evidence suggested that the environment had been modified to establish persistence, maintain privileged access, and enable remote administration capabilities.

Additional SIEM analysis was required to validate these findings through log correlation.

Investigation Phase 2 – SIEM Validation

Objective

After identifying suspicious host artifacts, I used Splunk to validate whether attacker activity could be observed through centralized log collection and event correlation.

The goal was to determine whether authentication activity, PowerShell execution, and suspicious downloads could be detected through SIEM telemetry.

Investigation Process

I reviewed Windows Security Logs, PowerShell Operational Logs, and Sysmon telemetry collected by Splunk.

The investigation focused on identifying authentication anomalies, suspicious command execution, and indicators of potential attacker activity.

Findings

Failed Authentication Attempts

Multiple failed login events (Event ID 4625) were identified during log analysis.

The frequency and pattern of these events were consistent with password guessing or brute-force activity.

PowerShell Execution Activity

PowerShell logs revealed the execution of encoded commands using:

```
powershell.exe -EncodedCommand
```

Encoded PowerShell commands are commonly reviewed during investigations because they may be used to conceal malicious activity.

Suspicious Download Activity

Additional review identified PowerShell download activity involving:

```
Invoke-WebRequest
```

This activity demonstrated how files could be retrieved from external sources and highlighted the importance of monitoring command execution through centralized logging.

Assessment

SIEM analysis successfully validated the host-based findings and provided additional visibility into authentication activity, PowerShell execution, and suspicious download behavior.

The correlation of host artifacts and SIEM telemetry strengthened confidence in the investigation findings and demonstrated the value of centralized log monitoring.

Skills Demonstrated

- Splunk log analysis
- Event correlation
- Authentication monitoring
- PowerShell investigation
- Threat detection
- SOC investigation workflow

SIEM Evidence & Validation

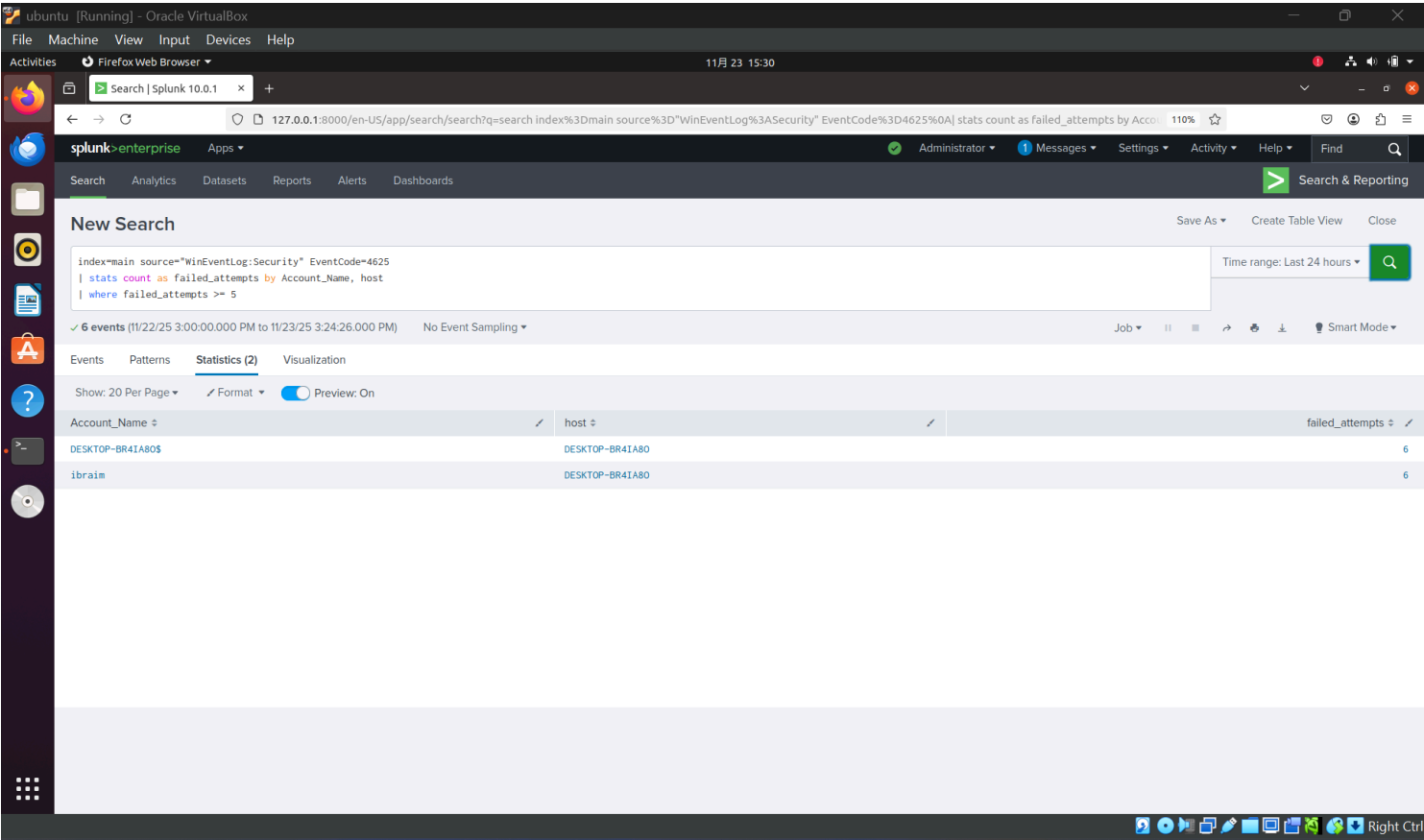
Evidence 1 – Failed Authentication Activity

While reviewing Windows Security logs in Splunk, I identified multiple failed authentication events (Event ID 4625).

The volume and frequency of these events were consistent with a brute-force or password guessing pattern.

This finding supported the initial hypothesis that unauthorized access activity may have occurred within the environment. The detection identified six failed authentication attempts originating from the monitored endpoint, exceeding the configured threshold and triggering further investigation.

Screenshot: Splunk detection showing multiple failed logon attempts (Event ID 4625)



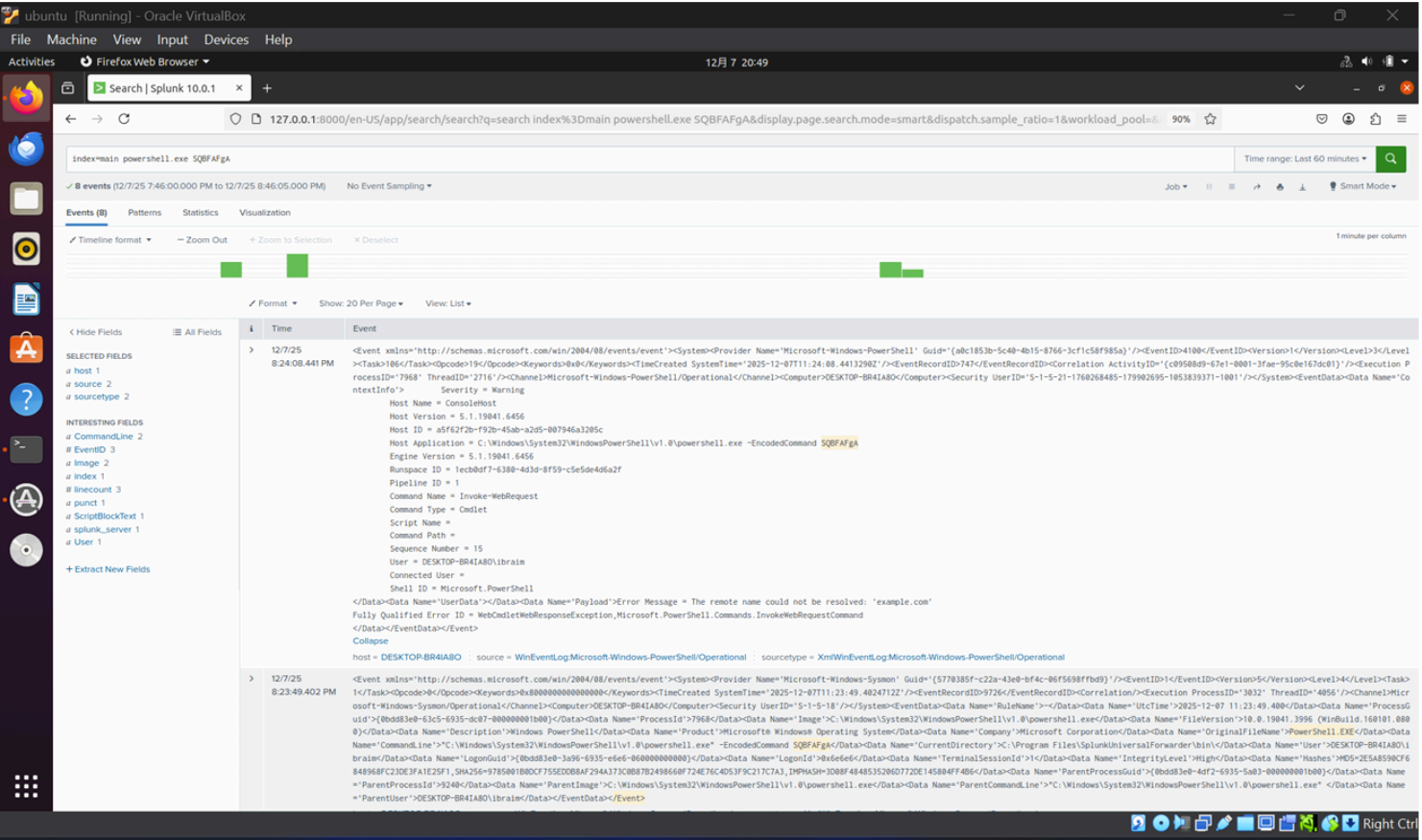
Evidence 2 – Encoded PowerShell Execution

Further investigation of PowerShell Operational logs revealed the execution of an encoded PowerShell command.

Encoded commands are frequently reviewed during incident investigations because they can be used to conceal malicious activity and bypass basic detection mechanisms.

The event provided additional evidence of suspicious command execution occurring on the endpoint.

Screenshot: PowerShell execution using -EncodedCommand



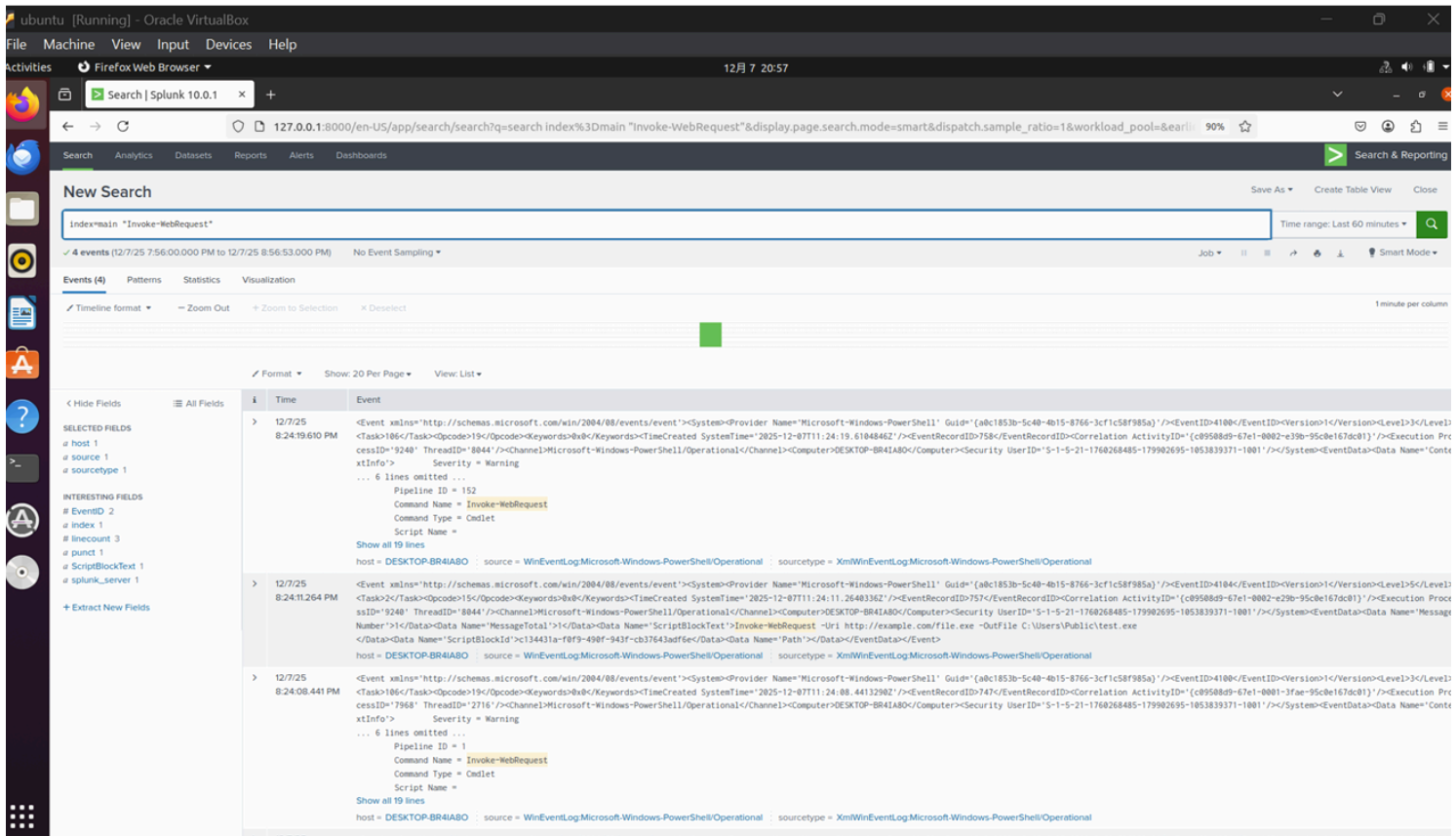
Evidence 3 – Suspicious Download Activity

Additional log analysis identified the use of Invoke-WebRequest , a PowerShell command capable of downloading files from external sources.

Although PowerShell download activity is not always malicious, it is commonly reviewed during investigations because attackers frequently use it to retrieve payloads or tools.

This finding further strengthened the hypothesis that the system had been used for suspicious activity.

Screenshot: Invoke-WebRequest activity observed in Splunk



Investigation Phase 3 – Linux Host Analysis

Evidence 4 – Suspicious Binary Execution

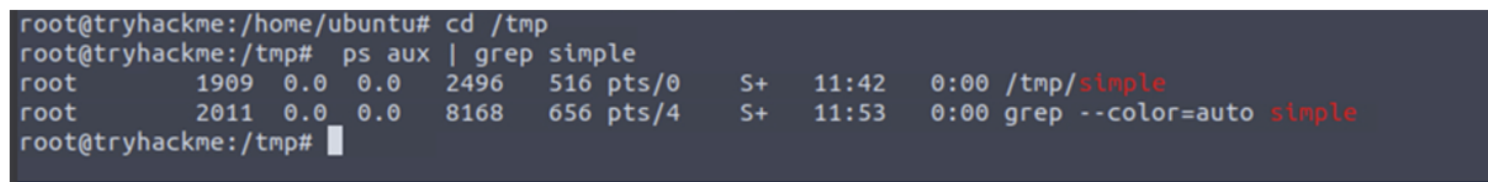
While reviewing running processes on the Linux host, I searched for unusual executables that did not appear to be part of normal system operations.

During this review, I identified a binary named `/tmp/simple` executing from a temporary directory.

Executables running from temporary locations are commonly investigated because attackers frequently use these directories to store and execute malicious payloads.

This finding supported the hypothesis that unauthorized activity had occurred on the Linux system.

Screenshot: Suspicious binary observed running on the Linux host



Evidence 5 – Unauthorized User Creation and Privilege Escalation

During the investigation, I reviewed local user accounts and privilege assignments to determine whether administrative access had been modified.

The analysis revealed the creation of a user account named **attacker**. Further review showed that the account had been granted full sudo privileges.

The combination of unauthorized account creation and elevated permissions suggested an attempt to establish persistent administrative access within the environment.

This finding represented a significant indicator of compromise and supported the theory that the attacker was attempting to maintain long-term control of the system.

Screenshot: Unauthorized account granted administrative privileges

```
ubuntu@tryhackme:~$ sudo su
root@tryhackme:/home/ubuntu# sudo useradd attacker -G sudo
root@tryhackme:/home/ubuntu# sudo passwd attacker
New password:
Retype new password:
passwd: password updated successfully
root@tryhackme:/home/ubuntu# echo "attacker ALL=(ALL:ALL) ALL" | sudo
c/sudoers
attacker ALL=(ALL:ALL) ALL
root@tryhackme:/home/ubuntu#
```

Evidence 6 – Persistence Through Cron Job

During the Linux host investigation, I reviewed scheduled tasks and cron configurations to identify mechanisms that could allow malicious activity to survive system reboots.

While examining the root user's crontab, I identified a scheduled task configured to execute the **/tmp/simple** binary automatically.

Persistence mechanisms such as cron jobs are frequently used by attackers to regain access or maintain execution after a system restart.

The discovery of this scheduled task provided additional evidence that the suspicious binary was intended to remain active within the environment over an extended period.

This finding strengthened the assessment that the attacker had established persistence on the Linux host.

Screenshot: Cron job configured to execute a suspicious binary at startup


```
root@ubuntu:~# cat /var/spool/cron/crontabs/root
# DO NOT EDIT THIS FILE - edit the master and reinstall.
# (/tmp/crontab.3EYGAE/crontab installed on Sat Dec 6 12:39:19 2025)
# (Cron version -- $Id: crontab.c,v 2.13 1994/01/17 03:20:37 vixie E
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
@reboot /tmp/simple
root@tryhackme:/var/spool/cron/crontabs#
```

Evidence 7 – Persistence Through Systemd Service

During the Linux host investigation, I continued reviewing mechanisms that could allow unauthorized activity to persist after system reboots.

As part of this review, I examined configured system services and identified a service named **suspicious.service** running on the host.

The service was configured to start automatically and remained active after system startup, indicating that it could be used to maintain execution without requiring user interaction.

Systemd services are commonly abused by attackers as a persistence technique because they allow malicious processes to launch automatically whenever the system starts.

The presence of both a suspicious service and previously identified cron-based persistence mechanisms suggested that multiple methods had been established to maintain access within the environment.

This finding further strengthened the assessment that the attacker had implemented long-term persistence on the Linux host.

Screenshot: Suspicious systemd service configured to run automatically


```
root@tryhackme:/var/spool/cron/crontabs# sudo systemctl status suspicious.service
● suspicious.service - Suspicious_Service
   Loaded: loaded (/etc/systemd/system/suspicious.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2025-12-06 12:47:52 UTC; 30s ago
     Main PID: 2301 (suspicious)
       Tasks: 1 (limit: 4671)
      Memory: 156.0K
     CGroup: /system.slice/suspicious.service
             └─2301 /home/activities/processes/suspicious

Dec 06 12:47:52 tryhackme systemd[1]: Started Suspicious_Service.
```

Investigation Timeline

How the Incident Unfolded

The investigation revealed evidence of activity affecting both Windows and Linux systems. By correlating SIEM telemetry with host-based artifacts, it was possible to reconstruct the likely sequence of events.

Stage	Observation	Assessment
1	Multiple failed authentication attempts detected in Splunk (Event ID 4625)	Possible password guessing or brute-force activity
2	Encoded PowerShell execution identified	Potential attacker execution and obfuscation
3	Invoke-WebRequest activity observed	Possible payload download attempt
4	Suspicious binary /tmp/simple discovered on Linux host	Unauthorized code execution
5	User account <code>attacker</code> created with elevated privileges	Privilege escalation
6	Cron-based persistence identified	Persistence mechanism established
7	Suspicious systemd service discovered	Additional persistence mechanism
8	Multiple persistence artifacts confirmed	Evidence of long-term access attempts

Investigation Summary

The investigation began with suspicious activity identified through Splunk telemetry on the Windows host.

As the analysis progressed, additional indicators revealed potential attacker activity including encoded PowerShell execution and suspicious download attempts.

Host-based analysis on the Linux system uncovered further evidence of compromise, including unauthorized user creation, privilege escalation, and multiple persistence mechanisms.

By correlating evidence across both systems, the investigation demonstrated a multi-stage compromise involving execution, privilege escalation, persistence, and remote access preparation.

Final Assessment

Investigation Findings

The investigation confirmed evidence of a multi-stage compromise affecting both Windows and Linux systems.

Key findings included:

- Unauthorized account creation
- Privilege escalation activity
- Multiple persistence mechanisms
- Encoded PowerShell execution
- Suspicious download activity
- Potential remote access tooling

- Credential access configuration changes

The combination of these findings suggests an attacker attempting to establish long-term access, maintain persistence, and execute additional payloads within the environment.

Key Indicators Observed

Indicator	Purpose
/tmp/simple	Suspicious binary execution
attacker	Unauthorized privileged account
suspicious.service	Persistence mechanism
Invoke-WebRequest	Suspicious file download activity
-EncodedCommand	PowerShell obfuscation
AnyDesk.exe	Remote access software
WDigest UseLogonCredential=1	Credential access configuration

MITRE ATT&CK Mapping

Throughout the investigation, several attacker techniques were identified and mapped to the MITRE ATT&CK framework.

Credential Access

T1110 – Brute Force

Multiple failed authentication attempts were identified through Windows Security Event ID 4625, indicating possible password guessing or brute-force activity against the Windows host.

Execution

T1059.001 – PowerShell

Encoded PowerShell commands were executed using the -EncodedCommand flag, demonstrating script-based execution and command obfuscation techniques commonly used by attackers.

Command and Control

T1105 – Ingress Tool Transfer

PowerShell Invoke-WebRequest activity indicated an attempt to download an external executable onto the Windows system, suggesting potential payload delivery.

Persistence

T1053.003 – Scheduled Task / Cron

A cron job was identified executing the suspicious binary /tmp/simple at system startup, providing persistence across reboots.

T1543.002 – Create or Modify System Process

A suspicious systemd service was configured to start automatically and remain active after reboot, representing an additional persistence mechanism.

Privilege Escalation

T1548.003 – Abuse Elevation Control Mechanism

The unauthorized user account attacker was granted full sudo privileges through modifications to the sudoers configuration, allowing elevated administrative access.

Credential Access

T1003 – OS Credential Dumping (Supporting Evidence)

WDigest-related registry modifications were identified, potentially exposing credentials in memory and increasing the risk of credential theft.

ATT&CK Techniques Summary

Technique ID	Technique
T1110	Brute Force
T1059.001	PowerShell
T1105	Ingress Tool Transfer
T1053.003	Scheduled Task / Cron
T1543.002	Create or Modify System Process
T1548.003	Abuse Elevation Control Mechanism
T1003	OS Credential Dumping

Investigation Summary

The MITRE ATT&CK mapping demonstrated attacker behavior across multiple phases of the attack lifecycle, including credential access, execution, persistence, privilege escalation, and command-and-control activity.

The combination of these techniques supported the assessment that the environment had experienced a multi-stage compromise rather than a single isolated event.

Response Actions

Based on the findings identified during the investigation, the following actions were recommended:

Immediate Actions

- Remove all identified persistence mechanisms, including cron jobs, systemd services, and startup artifacts.
- Disable and remove unauthorized user accounts.
- Delete suspicious binaries, scripts, and attacker-controlled tooling.
- Reset potentially compromised credentials.
- Review systems for additional unauthorized changes.

Monitoring Improvements

To improve future detection capabilities, additional monitoring and alerting recommendations were identified:

- Monitor repeated failed authentication attempts.
- Alert on encoded PowerShell execution.
- Monitor PowerShell download activity such as Invoke-WebRequest.
- Review persistence-related changes to services, startup locations, and scheduled tasks.
- Enable enhanced PowerShell logging and endpoint telemetry.

Security Enhancements

- Implement account lockout policies.
- Enable Multi-Factor Authentication (MFA) where possible.

- Perform regular reviews of privileged accounts.
- Conduct periodic threat hunting activities.
- Validate endpoint logging coverage across Windows and Linux systems.

Lessons Learned

This investigation demonstrated how suspicious activity identified through SIEM telemetry can be correlated with host-based evidence to uncover a broader compromise.

The investigation highlighted the importance of reviewing authentication events, PowerShell activity, persistence mechanisms, privilege escalation indicators, and remote access tooling when assessing potentially compromised systems.

By combining Windows telemetry collected in Splunk with Linux host analysis, it was possible to reconstruct the attack chain and identify multiple stages of attacker activity.

This project strengthened practical SOC analyst skills including log analysis, threat detection, incident investigation, evidence correlation, MITRE ATT&CK mapping, and incident documentation.

Conclusion

This case study demonstrates a structured SOC investigation workflow from initial detection through evidence collection, analysis, validation, and remediation planning.

The investigation successfully identified indicators of execution, privilege escalation, persistence, credential access, and potential command-and-control activity across both Windows and Linux systems.

Although performed within a controlled lab environment, the investigation reflects many of the techniques and workflows used during real-world SOC and incident response operations.